WEB SYSTEMS AND TECHNOLOGIES

Phase 3 – Frontend Development Based on Project Design

LAB REPORT

Uniformity: An E-Commerce platform for Pre-Loved School and Professional Uniforms

GROUP NAME: TechNic **Course & Block:** BSIT-2A		
MEMBERS	**ROLE**	**SIGNATURE**
Cipriano, Mcxyron B.	*Back-End Developer*	
Hernandez, Kurt Jushua S.	*Database Manager/Tester & Debugger*	
Registrado, Mharie Franz	*Tester & Debugger*	
Red, Paul Orlando B.	*Project, Database, GitHub Manager & Documentation Officer*	
Romano, Jay L.	*Front-End Developer*	

Submitted to:

Guillermo V. Red, Jr., DIT

Assistant Professor IV

I. Overview of Frontend Development

This laboratory report documents the Phase 3 frontend development of our Final Project —Uniformity, a web-based e-commerce platform for buying and selling pre-loved school and professional uniforms. Building on the Node.js / Express / MongoDB backend established in Phase 2, Now for this 3rd phase, we deliver a complete, responsive user interface using HTML5, CSS3, and vanilla JavaScript.

Tech Stack Used:

HTML5	Semantic page structure and accessible form markup
CSS3 / Custom Properties	Design tokens, layout, animations, and responsive breakpoints
Bootstrap 5.3	Grid system, utility classes, modals, and form validation styles
Bootstrap Icons 1.11	Icon set used throughout navigation, buttons, and badges
Google Fonts – Poppins	Primary typeface for all UI text

Pages Developed:

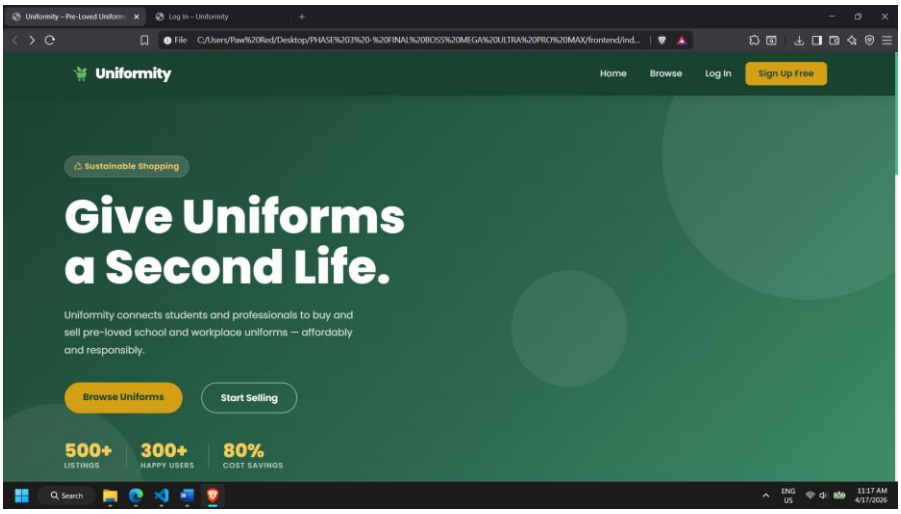
- ***index.html -*** Public landing / home page
- ***register.html -*** New user registration form
- ***login.html -*** User login form
- ***dashboard.html -*** Browse all uniform listings
- ***product-detail.html -*** Single product view with add-to-cart
- ***cart.html -*** Shopping cart with quantity controls
- ***checkout.html -*** Delivery address & order placement
- ***my-orders.html -*** Buyer order history
- ***my-order-details.html -*** Detailed single-order view
- ***my-listings.html -*** Seller product management table

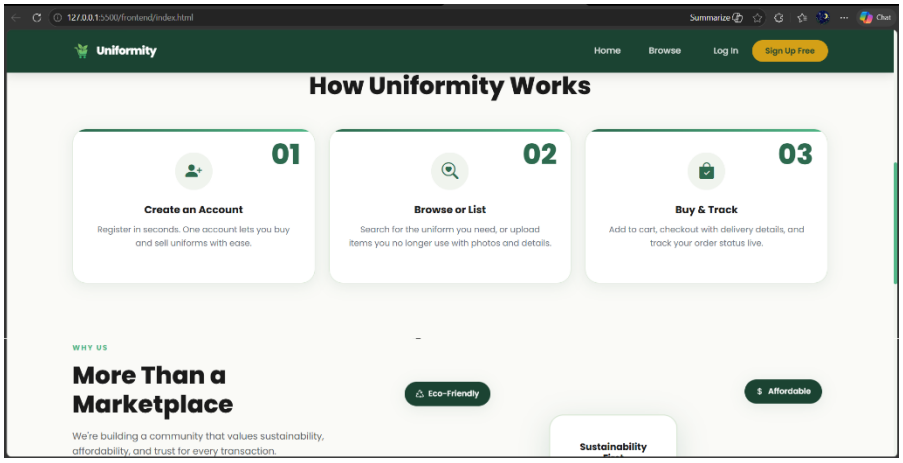
- ***add-listing.html -*** Create a new uniform listing with image upload
- ***profile.html -*** User profile view & edit modal

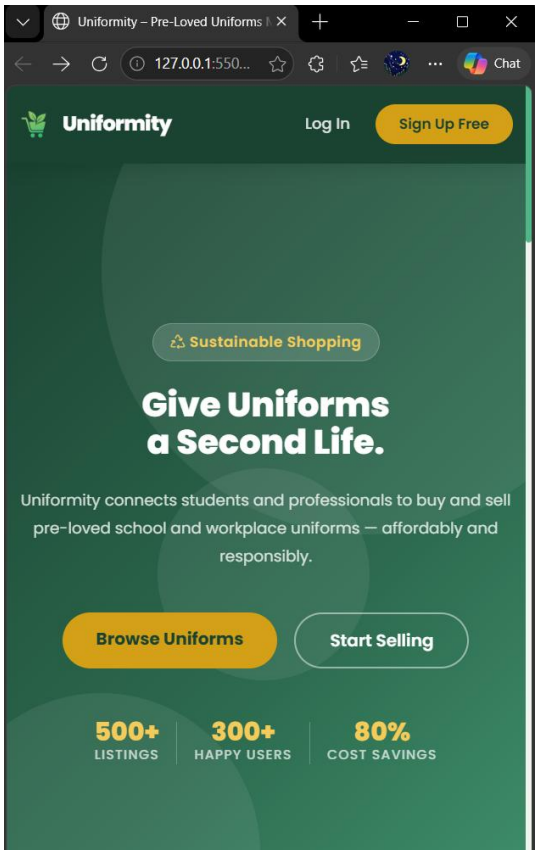
II. Frontend Pages

1. *index.html -* Public landing / home page

This page serves as the **entry point** for all actors (Buyers and Sellers) before authentication. It outlines the core system flow (List, Buy, Fulfill) represented in our System Architecture and routes users directly to the Authentication forms or the Product Dashboard.







Use Case Covered: Platform Entry & Navigation Gateway

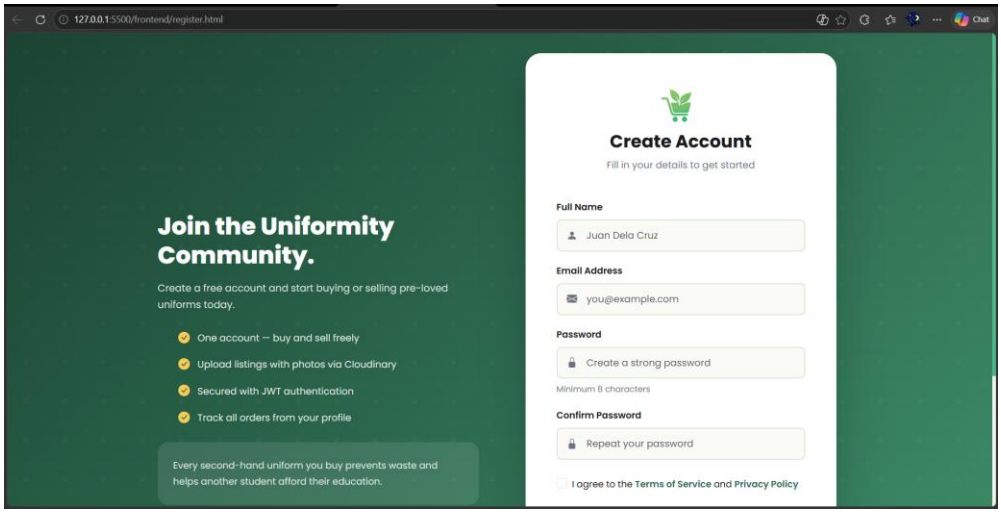
Mobile-First Layout & UI Adaptation: implemented a fully responsive design using Bootstrap 5's grid system combined with custom CSS media queries.

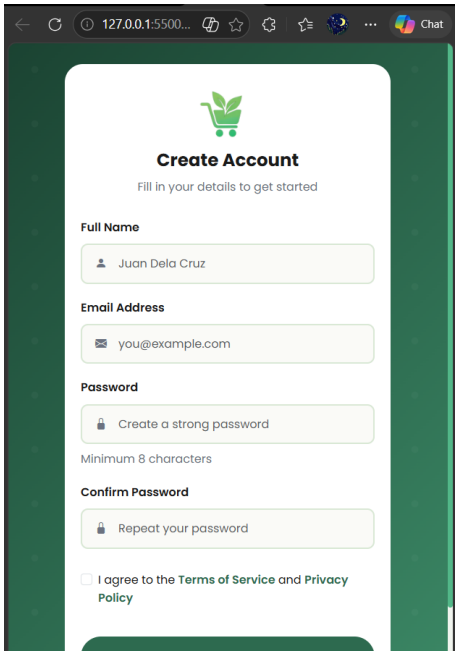
2. Authentication Pages

- ***register.html -*** new user registration form
- ***login.html -*** User login form

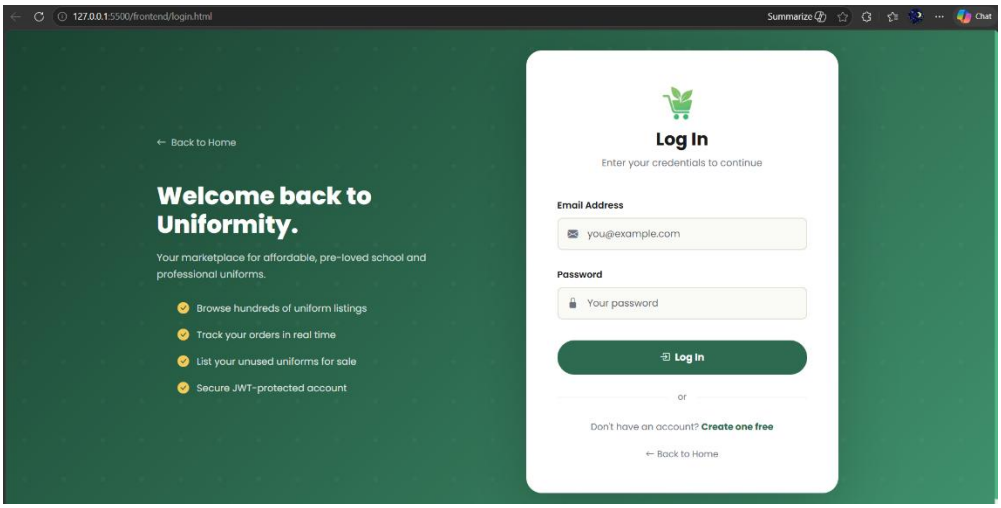
These pages provide the interface for user authentication. The **input fields (Name, Email, Password)** directly correspond to the string attributes defined in the USERS entity of our ERD. The form is set up with method="POST" and action="#" placeholders in preparation for JWT integration.

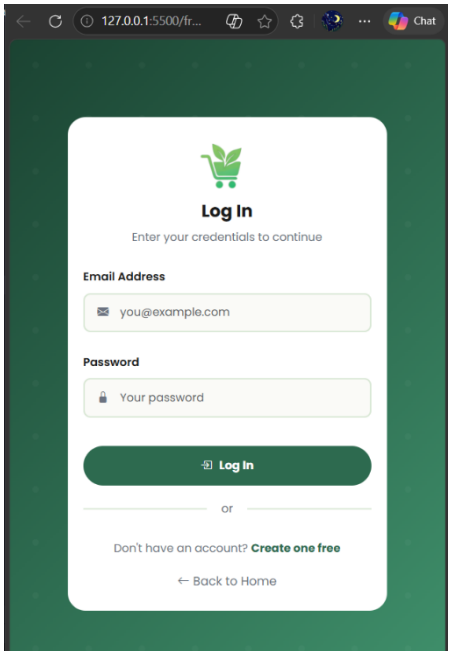
Use Case: Sign Up / Create account





Use Case: Login / Authenticate User



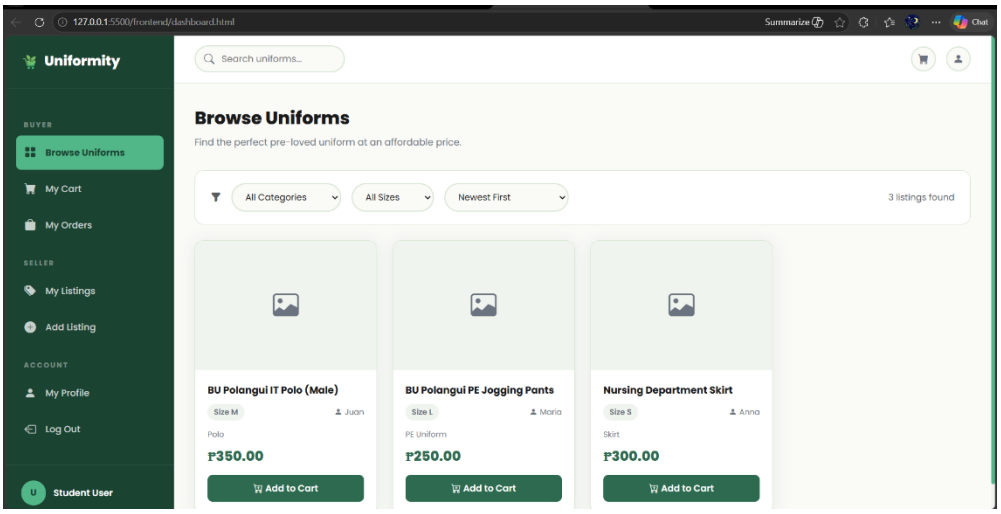


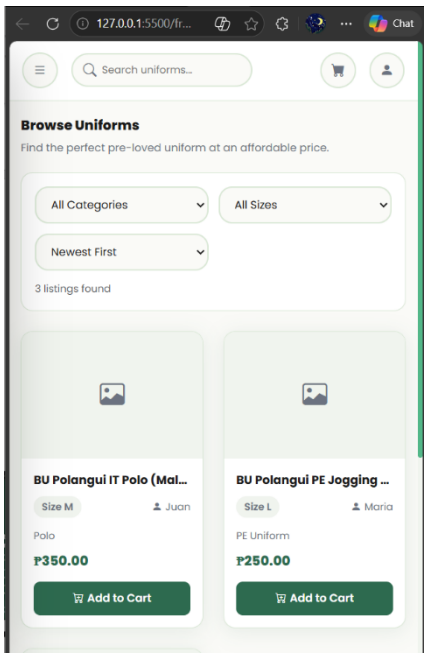
3. Buyer Flow: Browsing and Details

- ***dashboard.html -*** Browse all uniform listings
- ***product-detail.html -*** Single product view with add-to-cart

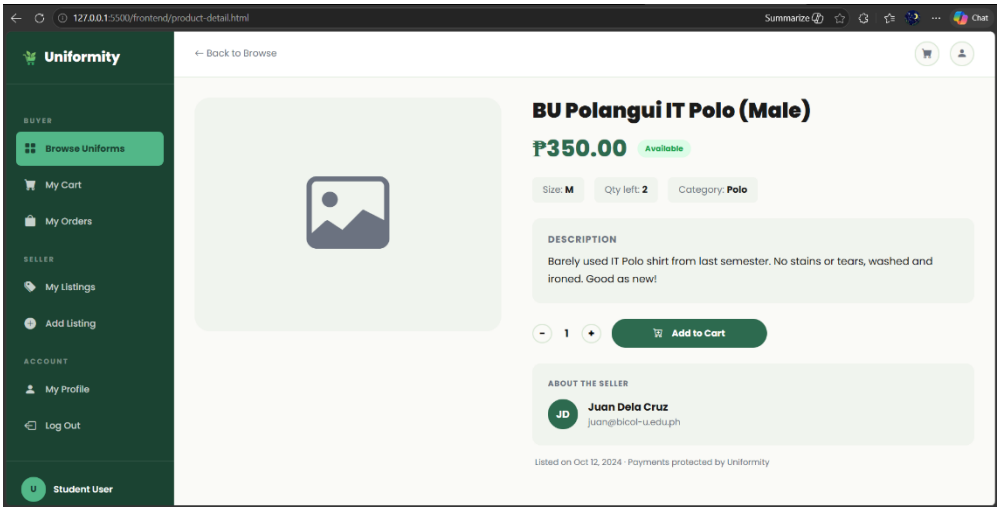
The dashboard simulates the retrieval of records from the **PRODUCTS table**, categorized via the **CATEGORIES table** constraint. The detail page reflects the specific attributes **(size, price, description, quantity, and image_url)** defined in the ERD.

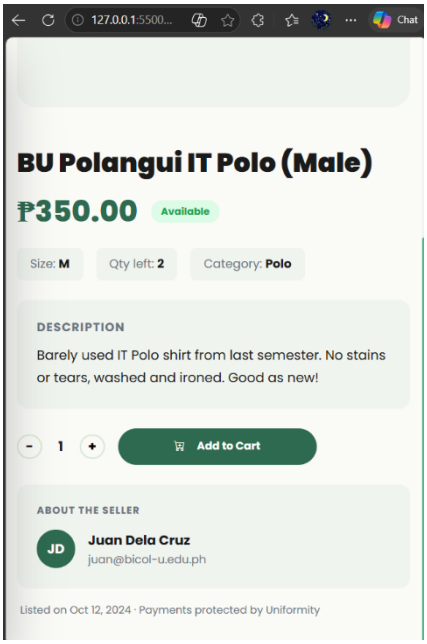
Use Case: Browse Uniforms (as Buyer) & Add to cart





Use Case: View Uniform Details; Add to cart



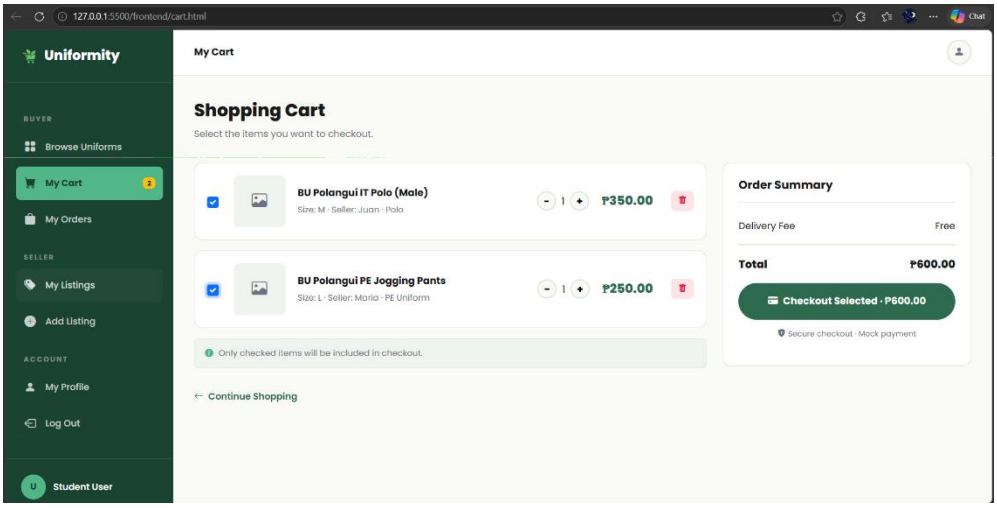


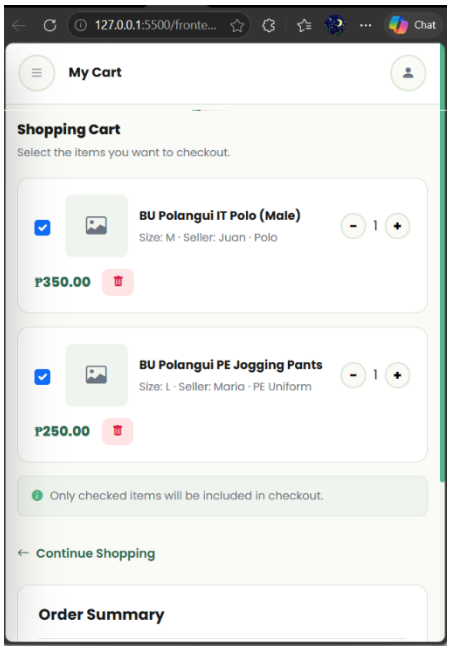
4. Buyer Flow: Cart and Checkout

- ***cart.html -*** Shopping cart with quantity controls
- ***checkout.html -*** Delivery address & order placement

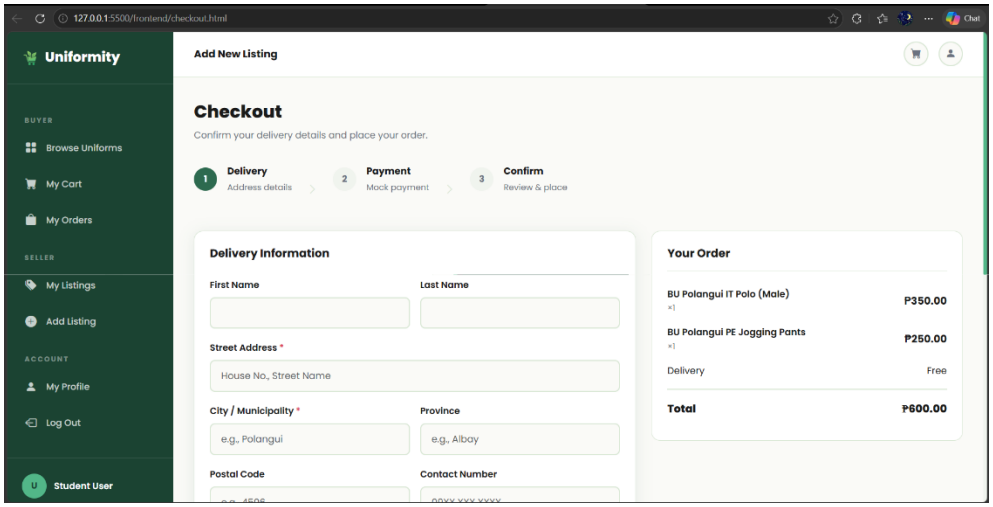
The cart page visualizes the **CART** and **CART_ITEMS** entities, linking a specific **user_id** to **product_ids**. The checkout form collects the **delivery_address** and calculates the **total_amount** required to populate the ORDERS table upon submission.

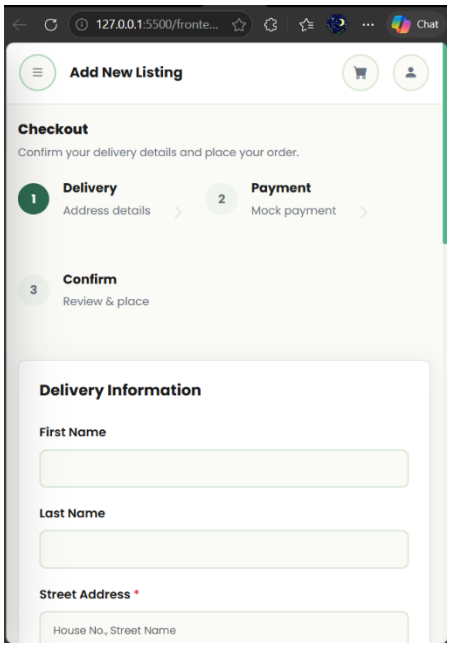
Use Case: Add to cart / Checkout





Use Case: Check out / Place Order



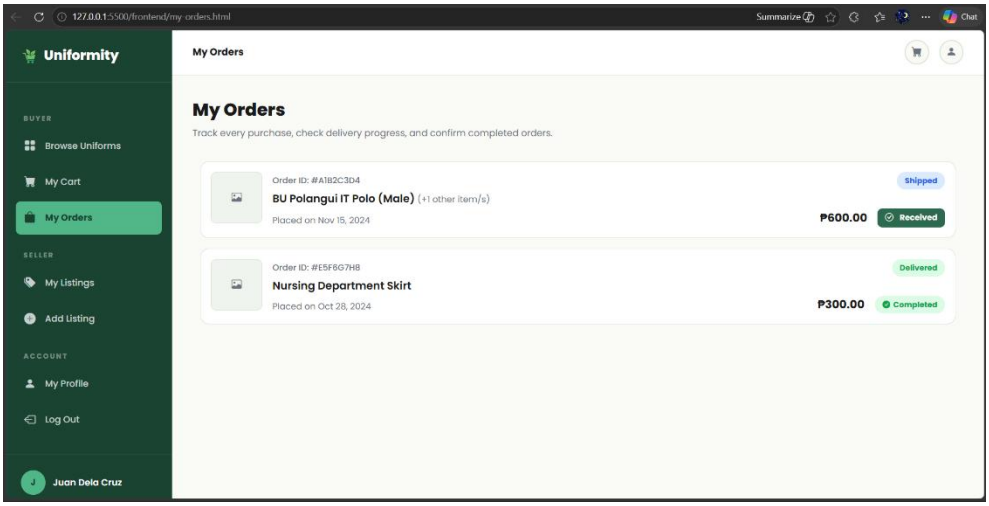


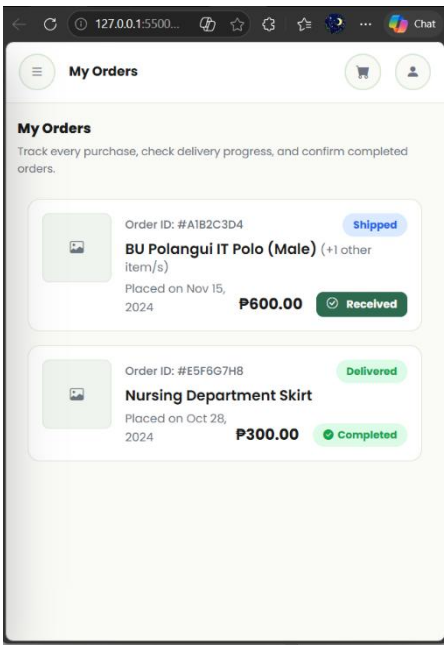
5. Order Tracking

- ***my-orders.html -*** Buyer order history
- ***my-order-details.html -*** Detailed single-order view

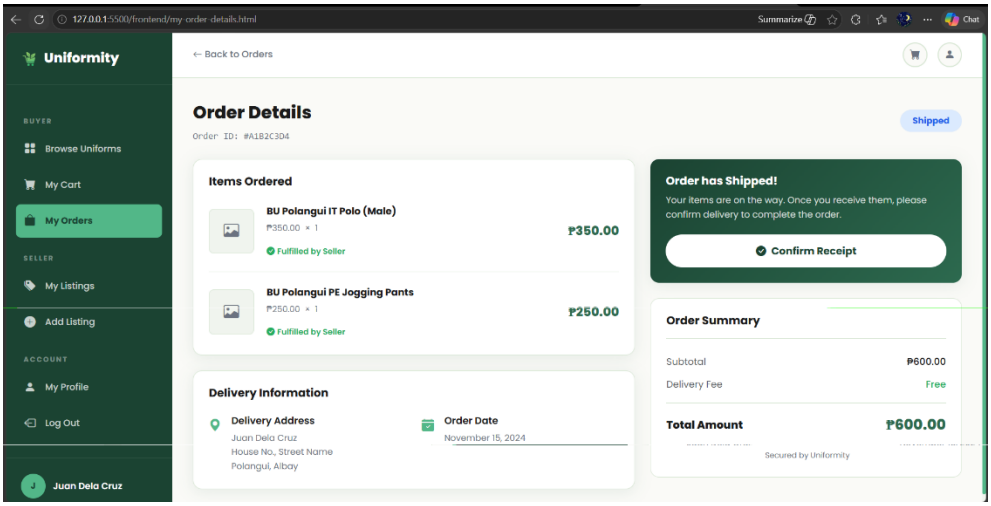
Displays the status of the transaction. The UI relies on the **status string attribute** from the **ORDERS table** (e.g., Pending, Shipped, Delivered) and groups **ORDER_ITEMS** by their associated **order_id.**

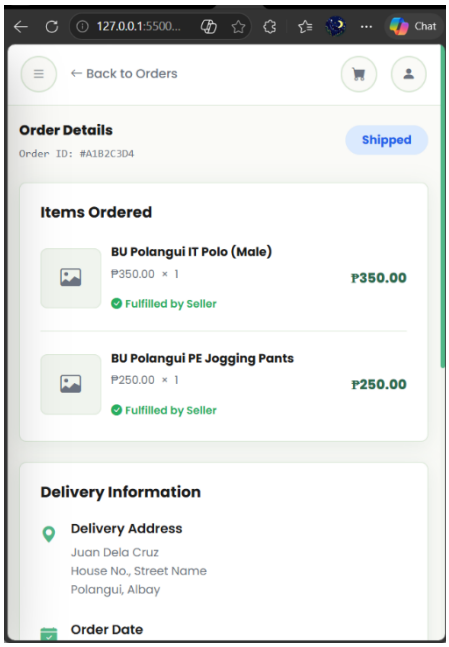
Use Case: Track Orders





Use Case: Track Orders



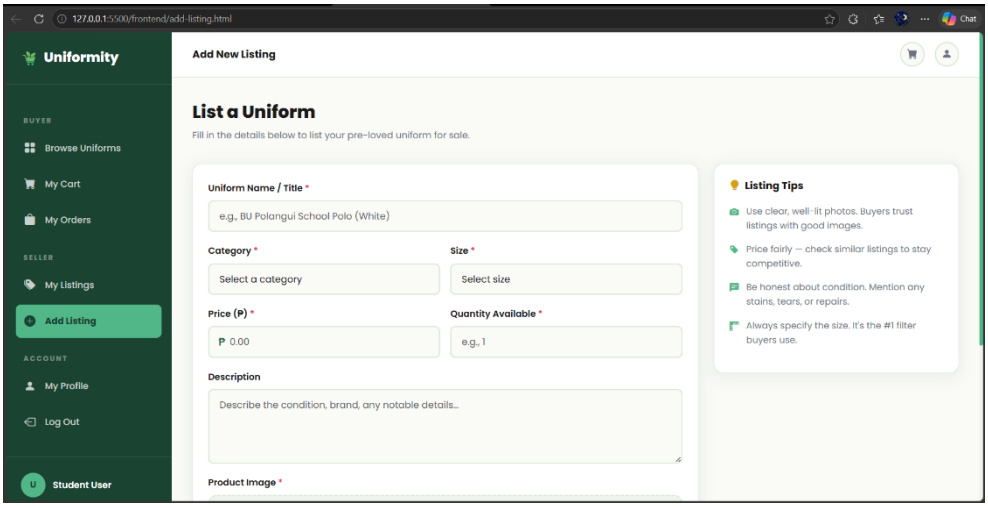


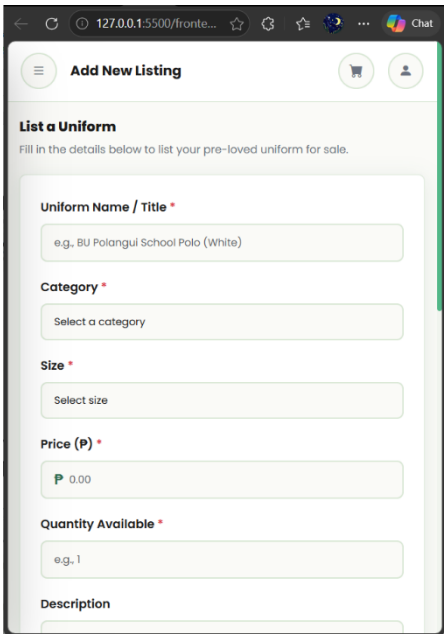
6. Seller Flow: Inventory and Fulfillment

- ***my-listings.html -*** Seller product management table
- ***add-listing.html -*** Create a new uniform listing with image upload

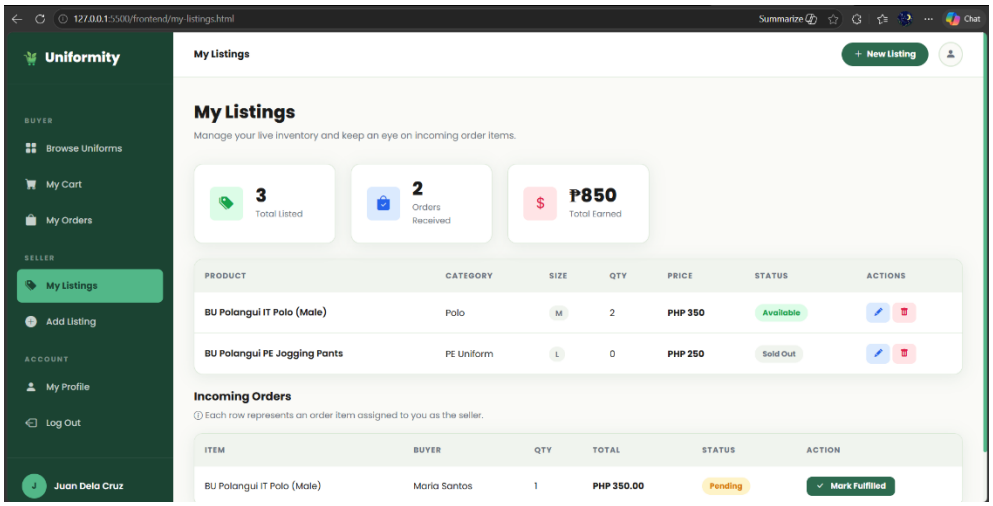
Represents the dual-role nature of our platform where a single User acts as a Seller. The Add Listing form uses **enctype="multipart/form-data"** to handle Cloudinary image uploads. The tables pull from the **PRODUCTS and ORDER_ITEMS entities** to allow sellers to manage stock and mark incoming items as fulfilled.

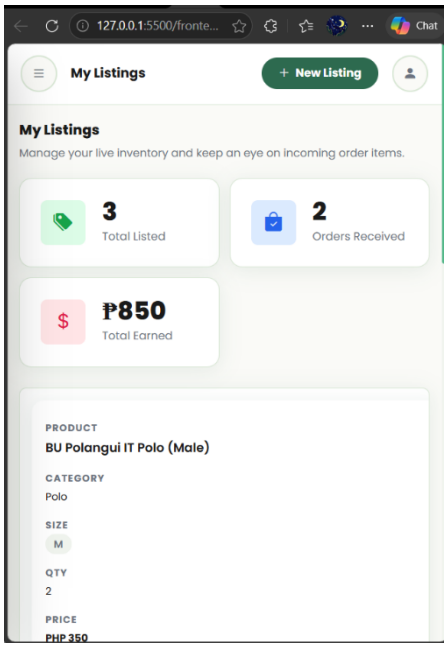
Use Case: List a uniform



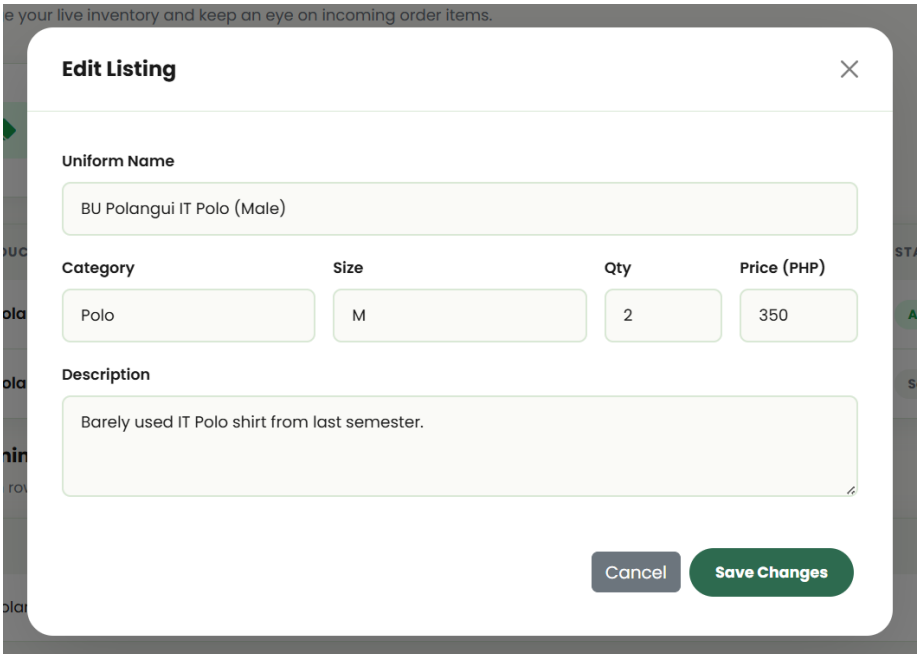


Use Case: Manage own listing & Fulfill Orders





Use Case: Manage own listing & Fulfill Orders

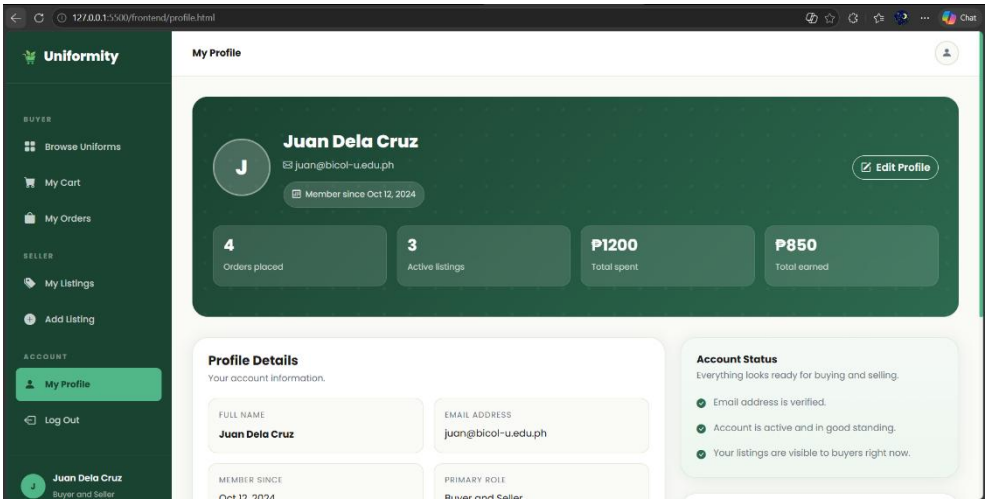


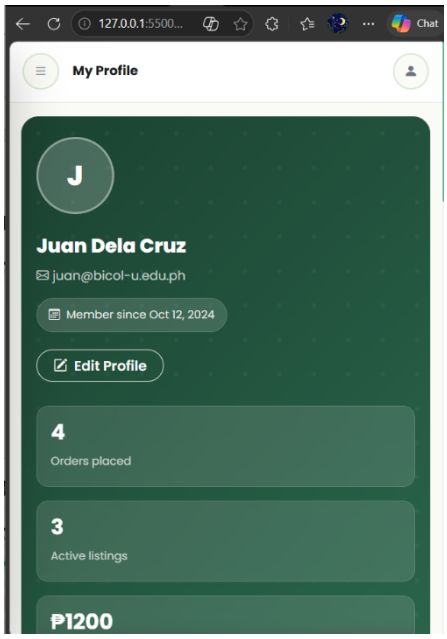
7. User Account Management

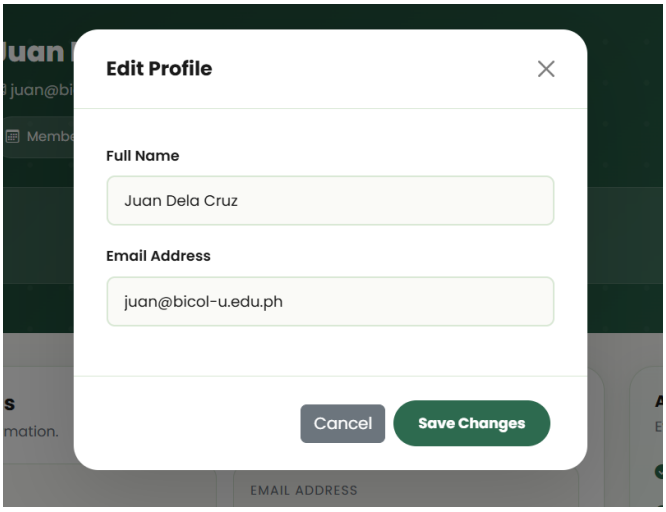
- ***profile.html -*** User profile view & edit modal

Serves as the central hub linking the User to their dual roles (Buyer and Seller pipelines). It uses Bootstrap cards to display mock aggregations of the user's spending (Orders) and earnings (Products sold).

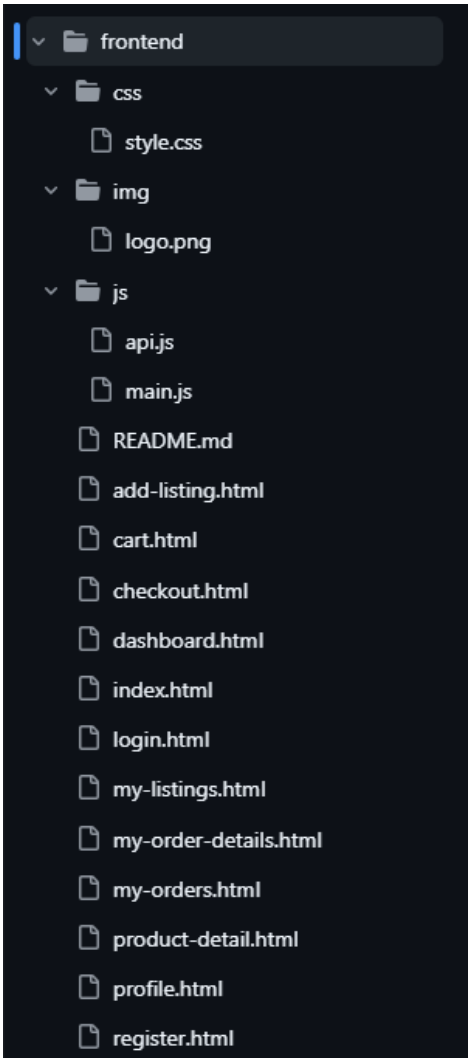
Use Case: Log in

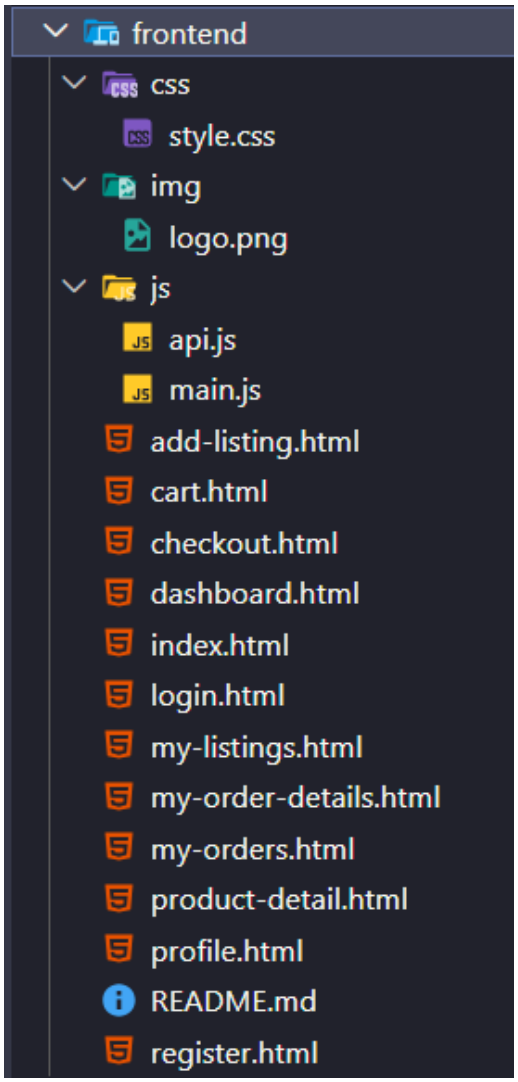




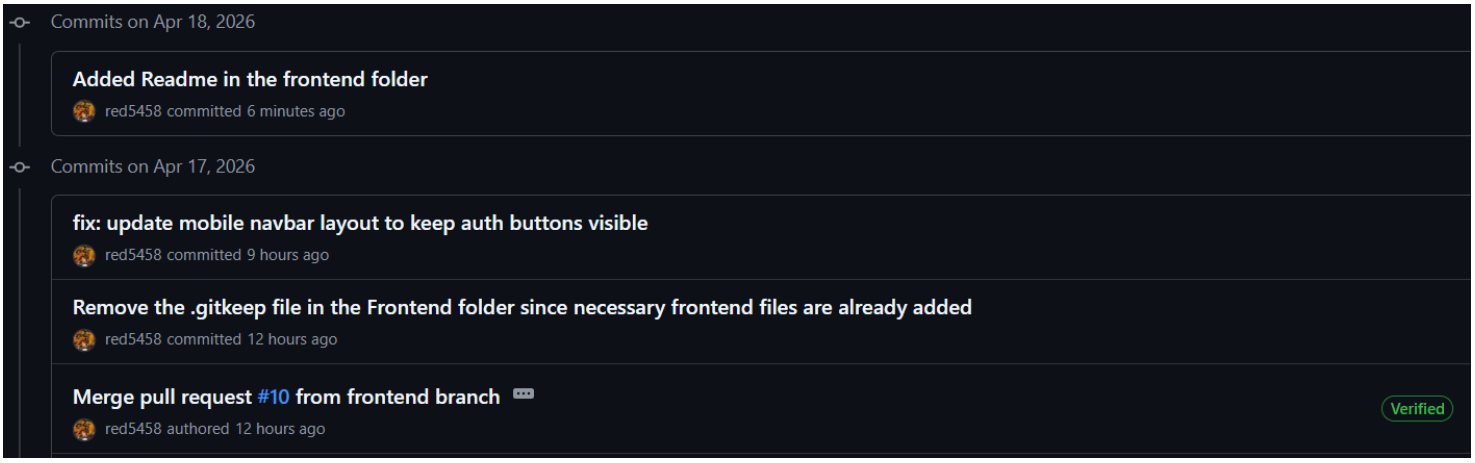


III. Screenshot of GitHub Repository Folder Structure





IV. GitHub Commits



V. Group Member Contributions (Individual Reflection)

- **Cipriano, Mcxyron —** Back-End Developer

Even if I had no direct frontend responsibilities assigned for this phase, I still assisted a little in building and refining the HTML, CSS, and Bootstrap templates based on the approved design documents. I also helped ensure that the UI elements were consistent with the project proposal from Phase 1 and properly reflected the planned system flow. In addition, I reviewed layout consistency and checked that the interface remained clean, functional, and aligned across all pages. I also contributed to making sure the overall design matched the expected user interactions and project requirements.

- **Hernandez, Kurt Jushua —** Database Manager / Tester & Debugger

I was not really assigned in any frontend responsibilities for this phase, but the manager also ask me to test the frontend user facing pages so I helped test the frontend against our planned data structures and noticed that some form field names did not match the Mongoose schema, which we corrected early. I also used our ERD from Phase 1 as a reference to ensure that form inputs properly corresponded to the correct database fields. In addition, I helped observe UI consistency and reported minor issues during testing to improve overall reliability.

- **Registrado, Mharie Franz —** Tester & Debugger

I took part in testing all 12 frontend pages to ensure everything worked as expected. During the process, I discovered a few issues such as the modals are not working which were later fixed. I also checked navigation flows, form validations, sidebar interactions, and overall responsiveness across pages. This experience helped me understand how important thorough UI testing is before backend integration is fully connected.

- **Red, Paul Orlando —** Project, GitHub, Database Manager, & Documentation Officer

I managed the GitHub workflow for Phase 3, including branch organization, repository structure, and maintaining a clean and consistent commit history across the team. I handled the final integration of all frontend pages, ensuring everything was properly completed, and committed. Even if it was not strictly part of my assigned responsibilities, I stepped in as a manager to help in developing the frontend pages, refine UI elements and fix minor bugs whenever needed to keep the frontend consistent and functional. I also made the full lab documentation report, updated the README, and coordinated task distribution to ensure the project stayed aligned and on track for submission.

- **Romano, Jay —** Frontend Developer

With the help of our Project manager, I was successfully build 12 fully designed pages in a single phase, It was challenging but rewarding. I handled the frontend development by creating consistent layouts using HTML, CSS, and Bootstrap to make sure all pages followed the same design system. I also learned how useful shared CSS variables in :root are, since defining colors, spacing, and styling once made global updates much easier. I also made sure the UI stayed consistent and aligned by checking and refining parts when needed. I also realized how important it is to design with the backend in mind, especially naming form fields to match API expectations.